

QUESTIONS - CASE STUDY 3 (PRACTICE):

MLA Citation:

Woodford, Teresa. "Case Study 3 (Practice): The Giant Magellan Telescope." *Physics Case Studies*. March 26, 2026. <https://www.physicscasestudies.org/GMT>

1. **Describe** how the development of the GMT (a scientific advancement) was intended to solve an economic problem in the Atacama Desert in Chile.

The Atacama Desert is an extreme environment with limited natural resources, which traditionally restricts the types of industries that can thrive there. The development of the GMT acts as a **scientific catalyst** for economic growth. By choosing this remote location for its atmospheric clarity, the project creates a high-demand "hub" for specialized labor.

Specifically, the GMT intended to solve the problem of limited local opportunity by providing **well-paying jobs** in engineering, high-tech maintenance, and construction. This shifts the local economy from potentially stagnant or subsistence-based to one integrated with international scientific funding. However, the application of this science also highlights an economic tension: while it brings jobs, the need to truck in water due to the desert's dryness drives up the **cost of living**, meaning the economic "solution" requires careful management to ensure the local population actually benefits from the increased cash flow.

2. **Analyze** any two scientific implications of using "segmented mirror," technology

Implication 1	Implication 2
The transition from monolithic to segmented mirrors has profound scientific implications for how we collect and process light: • Overcoming Structural Deformation (Gravity): In a traditional monolithic design, a mirror the size of the GMT (24.5m) would be so heavy that it would warp under its own weight. Scientifically, even a microscopic change in the concave curvature ruins the focus. By using seven	• Scaling the Aperture (Light-Gathering Power): The "light bucket" principle states that more surface area equals more photons. The implication of segmented technology is that it allows astronomers to bypass the physical limits of glass casting. By tiling segments, the GMT achieves a much larger effective aperture. This allows the telescope to resolve images of objects with extremely low magnitude (very faint), such as galaxies billions of

8.4m segments, the GMT maintains structural rigidity . Each segment is stiff enough to hold its precise shape, ensuring that the Law of Reflection is applied accurately across the entire surface so that light rays converge perfectly at the focal point	light-years away, which would be scientifically impossible to detect with a smaller, single-piece mirror.
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3. **Discuss** one positive and one negative implication of using "segmented mirror" technology considering one of the following factors: ethical OR cultural.

What factor did you choose - cultural or ethical?	
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One Positive	One Negative
When considering cultural factors, the use of segmented mirror technology at the GMT site presents a conflict between "global scientific culture" and "local indigenous/historical culture." • Positive Implication: The technology fosters International Cultural Collaboration. Because segmented mirrors are so complex to build and align (using Adaptive Optics), the project requires the brightest minds from different nations to work together. This creates a shared global culture of discovery and puts Chile at the center of humanity's search for its origins, potentially becoming a point of national and cultural pride for the Chilean people.	Negative Implication: There is a risk of Cultural Displacement or Heritage Loss. The Atacama Desert is home to important archaeological sites. The massive physical footprint required to house a segmented mirror array—including the support buildings and roads needed for such a large-scale engineering feat—may encroach upon or damage ancient sites. Culturally, this represents a trade-off where "modern scientific progress" is prioritized over the preservation of "ancestral history," potentially alienating local communities who value the land for its past rather than its view of the stars. Negative Ethical Implication: Resource Inequity and Exploitation. There is a major ethical concern regarding distributive justice in the Atacama Desert. The segmented mirror technology requires a massive infrastructure in a region of "extreme dryness." While the

The use of "segmented mirror" technology, while a massive leap for astronomy, creates a significant ethical tension between the **pursuit of universal knowledge** and the **well-being of local populations**.

- **Positive Ethical Implication: The Democratization of Knowledge.** Ethically, the segmented mirror design is a triumph of "shared humanity." Because a single 24.5-meter mirror is impossible to build, the segmented approach allows multiple nations to contribute resources and expertise to a single goal. This creates an ethical framework of **global cooperation**. By pooling 1 billion US dollars, the project ensures that the data collected—which could answer fundamental questions about the origin of life and the universe—is a collective human achievement rather than the property of a single wealthy nation. It fulfills the ethical responsibility of science to expand the boundaries of human understanding for the benefit of all.

project brings "well-paying jobs," it also necessitates trucking in vast amounts of water. Ethically, this creates a conflict: is it "right" to prioritize the cooling and maintenance of high-tech mirrors and the needs of an international scientific colony when water is a scarce, life-sustaining resource for the local population? By driving up living costs and consuming local resources, the project risks marginalizing the vulnerable local community in favor of an elite scientific mission.

MYP Command Terms that are used on Criterion D:

Summarize: Abstract a general theme or major points	Outline: Give a brief account or summary
Apply: Use knowledge and understanding in response to a given situation	Describe: Give a detailed account or picture of a situation, event, pattern, or process
Discuss: Offer a considered and balanced review that includes a range of arguments, factors, or hypotheses. Opinions or conclusions should be presented clearly and supported by appropriate evidence	Explain: Give a detailed account including reasons and causes
Evaluate: Make an appraisal by weighing up the strengths and limitations	Identify: Provide an answer from a number of possibilities.

Rubric: Criterion D: Reflecting on the Impacts of Science

Achievement Level	Level Descriptor
0	The student does not reach a standard identified by any of the descriptors below
1-2	The student is able to: state the ways in which science is applied and used to address a specific problem or issue state the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical apply scientific language to communicate understanding but does so with limited success - documents sources with limited success
3-4	The student is able to: outline the ways in which science is applied and used to address a specific problem or issue outline the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical sometimes apply scientific language to communicate understanding sometimes document sources completely
5-6	The student is able to: summarize the ways in which science is applied and used to address a specific problem or issue describe the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical usually apply scientific language to communicate understanding clearly and precisely usually document sources completely
7-8	The student is able to: describe the ways in which science is applied and used to address a specific problem or issue discuss and analyse the implications of using science and its application to solve a specific problem or issue, interacting with a factor: social, environmental, economic, cultural, or ethical consistently apply scientific language to communicate

	understanding clearly and precisely • document sources completely
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MLA Citation Quick Reference Guide

PART A. CITING IN-TEXT CITATIONS

1. **Citing with a known author:** (Lopez, 87) or (Lopez)
 - Use the last name of the author and page number
 - Use the last name of the author only if you don't have a page number)
2. **Citing with an unknown author:** ("Green Energy Use")
 - Use the name of the article in quotes inside a bracket

PART B. CITING FOR YOUR MLA WORKS CITED

1. Website with One Author

Format:

Last name, first name. Title of the article in quotation marks. The name of the website in italics.
Date of Use. Web Link.

Sample:

Gadbery, Elizabeth. "The Best Houseplants for Low Light." *The Spruce*, 14 Jan. 2026,
www.thespruce.com/best-low-light-houseplants-4774244.

2. Website with More Than One Author

Format:

Last name, first name and First name, Last name. Title of the article in quotation marks. The
name of the website in italics. Date of Use. Web Link.

Sample (Two Authors):

Miller, Julian, and Sarah Chen. "Climate Trends in the Pacific Northwest." *Environmental Watch*, 2 Feb. 2026, www.envirowatch.org/climate-trends-pnw.

3. Website with More Than Two Authors

Format:

Last name, first name. Title of the Website in quotation marks. The name of the website in italics. Date of Use. Web Link.

Gadbery, Elizabeth at al. "The Best Houseplants for Low Light." *The Spruce*, 14 Jan. 2026, www.thespruce.com/best-low-light-houseplants-4774244.

4. Website with No Author

Format:

Title of the Website in quotation marks. The name of the website in italics. Date of Use. Web Link.

Sample:

"History of the Golden Gate Bridge." *History.com*, A&E Television Networks, 15 May 2025, www.history.com/topics/landmarks/golden-gate-bridge.

5. YouTube Video or Video from Other Source

Format:

Creator's Name or Channel Name. Title of the Video in Italics. The name of the website in italics. Date of Use. Web Link.

Sample:

Physics Girl. "Why the Sky is Blue (And Why it's Sometimes Green)." *YouTube*, 10 Nov. 2024, www.youtube.com/watch?v=bcdefg12345.

6. AI-Created Work

Format:

Prompt in quotation marks followed by the word prompt. AI Tool. Date of Use. Web Link.

Sample:

"Explain the concept of quantum entanglement in simple terms" prompt. *Gemini*, 3 Flash version, Google, 11 Feb. 2026, gemini.google.com.

